MTH 201: Probability and Statistics

End Semester Exam

01/05/2024

Binomial E[X]

Var[X]

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No books, notes, or devices are allowed. Just a pen/ pencil and eraser. Institute rules will apply with regards to cheating. Show your steps. Don't prioritize simplifying products and sums, if you are short on time. You have about 120 minutes.

Question 1. 15 **marks** A bin has 5 balls. Each ball has a number associated with it that is drawn from the Normal(10,1) distribution. An experiment involves choosing two balls without replacement from the bin. The experimenter takes away a reward that is a sum of the numbers on the chosen balls. Answer the following questions.

(a) (7.5 marks) Derive the expected value of the reward.

(b) (7.5 marks) Suppose you are told that one of the balls in the bin has the number 8 associated with it. Derive the expected value of the reward.

Question 2. 25 **marks** You are asked to take part in an assessment that has ten questions. For each question you are provided five possible answers, marked as choices (a), (b), (c), (d), and (e), of which exactly one is the correct answer. You are expected to choose your answer for each question. You get a score of 1 for a correct answer and a score of -1 for an incorrect answer. Your total score is a sum of scores you obtain for the ten questions. Answer the following.

(a) (7.5 marks) Suppose in the assessment the correct answer for a question is assigned uniformly and randomly to one of the five choices. Further suppose that you pick an answer to a question uniformly and randomly from the given choices. Calculate your expected total score and variance of the total score.

(b) (5 marks) For the random assignment of the correct answer to a choice as above, assume that you choose choice (a) as the correct answer for every question. Calculate your expected total score and variance of the total score.

(a) (7.5 marks) Repeat parts (a) and (b) for when the assessment has 100 questions and you score $-1/10$ for a wrong answer and $1/10$ for a correct answer.

(b) (5 marks) Which question paper and strategy for choosing answers would you prefer and why?

Question 3. 40 **marks** Two tasks, indexed 1 and 2, must be processed by two CPUs. Either CPU takes time to process any task that is distributed as a uniform random variable over the interval $(0, 1)$ seconds. All processing times are independent of each other. You can assign tasks in the following two ways. Under assignment A, task 1 is assigned to CPU 1 and task 2 is assigned to CPU 2 at the same time. Tasks begin processing as soon as they are assigned. The processing of all tasks is considered complete when both task 1 and 2 have been processed.

Under assignment B, task assignment takes place in two stages. First, task 1 is assigned to both the CPUs. As soon as one of the CPUs finishes processing task 1, both CPUs are reset and task 2 is assigned

to both of them. The processing of both tasks is considered to finish as soon as one of the two CPUs finishes processing task 2.

We are interested in the time taken to process a task, which is the time during which a task is being actively processed by any CPU. We are also interested in the total time taken to process both the tasks, which is the time that elapses between the instant when processing starts and the instant when processing of both tasks has finished. Answer the following questions.

(a) (18 marks) Derive the distribution of the time taken to process task 1 and the distribution of the time taken to process task 2 under assignment A. Also derive the distribution of the total time it takes to process both the tasks.

(b) (12 marks) Derive the distribution of the time taken to process task 1 and the distribution of the time taken to process task 2 under assignment B.

(c) (10 marks) Consider the total time taken to process both the tasks. Assume that this total can be well approximated by a suitably chosen Gaussian random variable. Derive the probability that assignment A takes a longer total time than assignment B. [Hint: Use the property that any linear combination of Gaussian random variables is Gaussian and leave the probability in terms of the CDF Φ of a standard normal.]

Question 4. 20 **marks** A wrist watch estimates the number of steps taken by a user during a day. The watch uses two pedometer sensors to track the number of steps and displays the mean of the two measurements to the user. For pedometer i, $i = 1, 2$, the number of steps is modeled by $S_i = s + W_i$, where s is the true (unknown) number of steps taken by a user and W_i is a random variable distributed as Gaussian(0,1). Assume that W_1 and W_2 are independent. Suppose at the end of a day S_1 takes a value of 100 and S_2 takes a value of 110. Derive a confidence interval estimate for the true number of steps s with confidence $1 - \alpha = 0.95$, using your measurements, in the following two ways. [Hint: Useful facts may come handy.]

(a) Using only the Chebyshev's inequality.

(b) Exploiting the knowledge of the distribution of W_i.

USEFUL FACTS

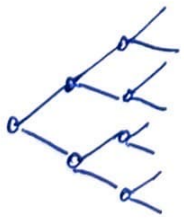
The PDF of a random variable X that is uniformly distributed over the interval (a, b), $a < b$, is given by $f_X(x) = 1/(b - a), x \in (a, b)$ and $f_X(x) = 0$ otherwise.

Fact: For any random variables X and Y, the variance of their sum is given by Variance$[X + Y]$ = Variance$[X]$ + Variance$[Y]$ + 2 Covariance$[X, Y]$.

Fact: The MGF of a Gaussian random variable X with mean μ and variance σ^2 is $\phi_X(s) = e^{s\mu + s^2\sigma^2/2}$.

Fact: A Gaussian random variable takes a value in the interval $(\mu - \sigma, \mu + \sigma)$ with probability 0.67, a value in $(\mu - 2\sigma, \mu + 2\sigma)$ with probability 0.95 and a value in $(\mu - 3\sigma, \mu + 3\sigma)$ with probability 0.99.

Fact: The Chebyshev's inequality states that $P[|X - E[X]| > c] \leq \text{Var}[X]/c^2$, $c > 0$.



$\frac{1}{10}$ $\frac{1}{10}$

$(1, -1)$

e^{2x^2}

$\cancel{2x\, e^{x^2}}$

$\cancel{2x}$